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Original research

Interrogating cortical representations in elite athletes with persistent posterior thigh pain – New targets for intervention?

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ABSTRACT

Objectives: Hamstring injuries in athletes can lead to significant time away from competition as a result of persistent posterior thigh pain. These cases are often difficult to treat as the state of the tissues alone cannot explain symptoms. In non-athletic populations with persistent pain, disruptions to tactile, proprioceptive, and spatial cortical representations exist, which has led to promising brain-based treatments. Here, we explored whether athletes with persistent posterior thigh pain also display impairments in these cortical representations.
Design: Cross-sectional study.
Methods: Fourteen male professional athletes with persistent posterior thigh pain ('Patients') and 14 pain-free age, sport, body mass index and level-matched controls ('Controls') participated. The tactile cortical representation was assessed using two-point discrimination (TPD) threshold and accuracy of tactile localisation; the proprioceptive cortical representation was assessed using a left/right judgement task; spatial processing was assessed using an auditory detection task.
Results: TPD thresholds were similar for Patients and Controls ($p = 0.70$). Patients were less accurate at localising tactile stimuli delivered to their affected leg, slower to make left/right judgements when the lower limb image corresponded to the side of their affected leg, and less accurate at detecting auditory stimuli delivered near their affected leg, when compared to their healthy leg or to the leg of Controls ($p < 0.01$ for all).
Conclusions: Leg-specific tactile, proprioceptive, and spatial processing deficits exist in athletes with persistent posterior thigh pain. That these processing deficits exist despite rehabilitation and normal tissue healing time suggests they may play a role in the persistence of posterior thigh pain.
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Practical implications

- This study shows that tactile, proprioceptive, and spatial processing deficits exist in elite athletes with persistent posterior thigh pain that are specific to the affected leg.
- Given that other studies have also demonstrated similar impairments in chronic pain populations, the current study raises the possibility that the disruptions observed may reflect a contributor to poor recovery, a possibility that seems worthy of further investigation
- Understanding the implications of these impairments for recovery, and risk of re-injury, following an acute hamstring strain appears warranted.

1. Introduction

Hamstring strains are the most prevalent injury in a number of sports, including Australian rules football, rugby union, soccer, and sprinting.[15] These injuries are characterised by posterior thigh pain, weakness and loss of function of the injured leg, and disruption of the hamstring muscle.[40] Time to return to sport after hamstring strains can be varied and difficult to predict,[15] but have been documented to take anywhere from a few days[48] to 104 weeks,[1] or in some cases have forced involuntary retirement from sport.[24] While several tissue-based factors (e.g., grade of injury) contribute to this time away from sport,[1] they do not always explain the duration of time lost.

Posterior thigh pain is a key reason for delayed return to competition. One relevant non-tissue related factor that may contribute to persisting posterior thigh pain is disruption in how the body and

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space surrounding it is represented in the brain (i.e., the cortical representation). Cortical representations are considered networks of brain cells that are distributed across multiple brain areas, each defined according to its relationship with a given output (e.g., a feeling, movement or behaviour), also termed 'neurotags'.[7,13] This concept of neurotags can be thought of as subserving brain function, including our motor and sensory systems and the mapping of stimuli into space. That is, the brain provides spatial coordinates for a stimulus' location and the closer the stimulus is to the body, the more likely it is to be processed in a different way.[8] To this end, one can conceptualise the various spatial fields immediately surrounding the body as 'peripersonal space'.[6,46] The spatial coordinates of peripersonal space are thought to be represented relative to multiple frames of reference, for example limb based and body midline based.[33] Multiple and precise representations of peripersonal space allows us to optimise protective strategies according to whether, and how, they are penetrating peripersonal space,[6] and to allow for interaction with the immediate surrounding environment. These cortical representations can be interrogated by using behavioural tasks that are believed to indirectly assess their integrity.[33] Tactile representation can be assessed by tactile acuity tasks (e.g., two-point discrimination and tactile localisation)[10] – loss of tactile acuity in some pain states relates to the extent of reorganisation in cortical representations, as assessed by brain imaging.[16,17] Proprioceptive representations can be assessed by a task in which a judgement is made as to whether an image of a body part belongs to the left or right side of the body. The task requires one to mentally manoeuvre their own body part from its current position to match the position of the body part displayed in the image–but without *actually* moving.[42] This task activates similar cortical networks as those activated when performing the actual movement.[41] Peripersonal spatial representation is thought to provide information about the position of objects or cues in the surrounding environment with respect to the body[33] and can be assessed for example by the localisation of auditory cues in the near-body space, the findings of which are important in the assessment of neglect-like disorders.[43]

Tactile acuity, left/right judgement, and spatial processing deficits exist in non-athletic populations with persistent pain (e.g., complex regional pain syndrome [CRPS]).[33] As such, these impairments have been attributed to disruptions in cortical representations, which has led to the development of 'brain-based' treatments that appear to both reduce pain and normalise cortical representations.[35] However, whether these deficits exist in elite athletes with persistent pain is unknown. Understanding the integrity of cortical representations in elite athletes with persistent posterior thigh pain may help explain persisting pain after hamstring injury.

This study assessed the integrity of cortical representations (by using a series of behavioural tasks that are thought to indirectly examine the tactile, proprioceptive, and spatial cortical representations) in elite athletes with persistent posterior thigh pain following a hamstring injury. Based on previous research in non-athletic populations,[5,10] we hypothesised that tactile, proprioceptive, and spatial processing deficits would be observed for the affected limb but not the unaffected limb or to the matching limb of healthy athlete controls.

2. Methods

The University of South Australia Human Research Ethics Committee approved the study. Data were collected as part of clinical assessments between 2007 and 2017 by GLM.

A convenience sample of 14 male professional athletes, participating in the highest level national competition in their sport, with persistent posterior thigh pain were recruited (Patients). Fourteen pain-free, age, sport, competition-level, and body mass index (BMI) matched controls were recruited via open invitations from within the same team, with the exception of one tennis player, squash player, and track sprinter (Controls). For the tennis and squash players, a Control participant was obtained via open invitation from athletes within 40 places on official world rankings in each sport. For the sprinter, a Control participant was obtained from the national sprinting training squad. Where more than one Control participant volunteered, one was randomly selected.

Patients had received a diagnosis of hamstring injury, which had (with the exception of the track sprinter) been corroborated with bilateral ultrasound examination within three days of the injury with a categorisation of Grade 1 (minor), 2 (partial tear), or 3 (complete or near complete) hamstring injury. They had then experienced posterior thigh pain, defined as pain (of any severity) that occurred during activity (i.e., contraction and/or stretch of the hamstring muscle belly) anywhere in the posterior thigh region, between the ischial tuberosity and distal insertions of the hamstring muscles.[21] Their posterior thigh pain was deemed persistent if their pain (of any severity) had continued on most days beyond the predicted recovery time,[45] based on the ultrasound examination and the opinion of the team physiotherapist and doctor. Patients received rehabilitation according to club or athlete-specific protocol. Aside from sport-specific components, rehabilitation involved broadly similar acute management, including: graded hamstring reactivation, progressive hamstring strengthening, running, and return to sport exposure protocols. Control participants were athletes who did not report any posterior thigh pain or hamstring injury. All control participants, except the track sprinter, underwent blinded ultrasound assessment to determine if the presence of ultrasound evidence of hamstring injury existed in the absence of pain. The Control track sprinter did not undergo ultrasound assessment because the Patient track sprinter did not.

Demographic data were collected for all participants, including: age (years), height (cm), weight (kg), mood (0–10 NRS), self-rated running intensity maximum (%), and strength of the affected leg (reported as a proportion of strength of the healthy leg on isotonic testing). Strength testing involved open chain isokinetic testing of knee flexion in standing, hip extension in standing and hip extension with knee flexion in lying.[28,50] In addition, 'Nordics', which involve forward tilting from the knee while in kneeling, were undertaken.[30] Additional data relating to injury were collected, including: Grade of initial injury, mechanism of injury, and weeks since injury. In addition, all participants were asked: "How confident are you in your hamstring?" Answered using a scale of 0–10, where 0 is "not at all confident" and 10 is "100% completely confident" and "How does your leg feel? For example, does it feel its normal size, shape, weight, temperature, etc.?". The adjectives participants used to describe their posterior thigh were recorded. Once participants had finished describing how their leg felt, they were given a list of additional adjectives, and asked to select the adjective(s) that they believed were relevant to their situation. The above demographic and injury related data are described in Supplementary 1.

To assess the integrity of tactile, proprioceptive, and spatial cortical representations, tactile acuity (two-point discrimination and tactile localisation), motor imagery (left/right judgement), and spatial processing (auditory stimulus detection) assessments were conducted, respectively. All tests were undertaken by an examiner who was blinded to group and blinded to injury side in the patient group.

Assessment of two-point discrimination (TPD) followed the method described elsewhere.[34] Briefly, mechanical sliding callipers with a 1 mm precision were applied to the participant's posterior

thigh until skin blanching occurred. The assessment commenced at 0 mm between the two points and gradually increased until the participant discerned two points. A series of paired ascending and descending assessments were conducted on each side (4 tests each side, 8 tests total) and the average of the 4 assessments on each side was used to determine TPD threshold. Data points for tactile acuity that were greater than 2 SD from the participant's overall mean from the 8 tests were considered to reflect a lapse in attention and were removed.[10] Studies have shown that TPD measurements taken with callipers have high reliability,[9] and concurrent validity with other tactile measures.[22]

The tactile localisation task was conducted using an iPad with the participants in relaxed standing. They held an iPad in front of them that displayed a photograph of the posterior view of their legs (from pelvis to calves). There were 25 trials. Each trial consisted of the researcher touching the participant's posterior thigh with a cotton tip and the participant marking with a stylus on the iPad photograph where they perceived the stimulus. The location of each stimulus was marked on a template used by the assessor. Prior testing showed that the trained assessor stimulated the participant within 4 mm of the target location. The order of locations stimulated was randomised. For each trial, the researcher delivered the stimulus and the participant was given 5 s to respond. An inter-stimulus interval of 4 s–8 s was randomly applied by the assessor. The primary outcome was Error: the distance (millimetres) between the true location of the stimulus and the location of the stimulus reported by the participant. Average error for the 25 trials was used for analysis. Studies have shown that tactile localisation tasks have high discriminant validity for the hands (for example[23]).

Left/right judgements were assessed using an in-house programme (participants 1–8 in both groups) or the *Recognise* programme (NOIgroup, Adelaide, Australia) (participants 9–14 in both groups) using a protocol described elsewhere,[36] with some minor differences. In short, participants were shown either a right or left limb (hand or foot) and were asked to identify whether the limb belonged to the right or left side of the body. Two practice trials – 40 hand images then 40 feet images – were performed. Images were randomly selected from two libraries, each containing 30 images (15 right-sided images and 15 mirrored reflections) of either hands or feet. Participants were given a 5-min break. Two trials were then conducted where participants viewed 60 images of the upper limb and 60 images of the lower limb, presented in random order, with body side and image rotation also randomised. For each trial, images were counterbalanced and randomised such that all 15 images were shown 4 times. That is, all participants received the same images, but in different orders and rotations. Average response time (RT) for correct responses for each side were analysed. Response times lower than 0.8 s, or greater than 5 s, were considered to reflect a lapse in attention and were removed.[36] Studies have shown left/right judgement tasks have high reliability (for example[49]) and are a validated measure of implicit motor imagery (for example[5]).

The auditory stimulus detection task involved detecting the location of an auditory 'click' in the space surrounding the leg. Participants stood relaxed with their eyes closed. Auditory stimuli were delivered approximately 4 cm in front of or behind either thigh using a mechanical clicker. Twenty mechanical clicks were delivered around each leg. The participant responded to each stimulus by pointing to the location in space where they perceived the stimulus to have occurred. Participants were given 5 s to make a response. Inter-stimulus interval was between 2 s and 5 s. The primary outcome was Detection: the proportion of trials that were detected and located on the correct side of the body midline, expressed as a percentage of the total number of trials. Whether the participant pointed to the correct side of the midline was judged by the assessor. Auditory stimulus detection tasks are considered a valid measure of auditory spatial processing.[18]

The analysis plan was prospectively published on Open Science Framework (protocol number: 0000031670). Data were analysed using SPSS (version 23 IBM Corp, Armonk, NY, USA). Due to the exploratory nature of the study, no corrections were applied for multiple analyses to minimise the risk of Type 2 error.[3] Normality was assessed visually (Q–Q plots) and whether 95% of data fell within two standard deviations (SDs) of the mean. Skewness was assessed using a histogram; skewness between −1.5 and +1.5 was considered roughly symmetric. Variance was considered homogenous if the quotient in SDs between groups was between 0.67 and 1.5. Statistical significance was set at $p < 0.05$.

TPD, tactile localisation and response times were compared between Patients and Controls using independent *t*-tests, and between the affected and unaffected legs of Patients using paired *t*-tests. If normality was violated, the non-parametric equivalent was used. To assess spatial processing, median and interquartile ranges for the accuracy in auditory stimulus detection were to be compared using Wilcoxon signed-rank tests, but were not, which was a deviation from the protocol (see below). The analysis plan and results for the secondary objectives, as stated in the original protocol, and for the reliability assessments for tactile localisation, left/right judgements, and auditory stimulus detection, are detailed in Supplementary 2 and Supplementary 3, respectively.

3. Results

Ultrasound evidence of hamstring damage was observed on the affected leg of all 13 of those Patients for whom ultrasound was available. Ultrasound evidence of a Grade 1 hamstring injury was observed on the unaffected leg of 4 Patients, and on the tested leg of 4 Controls. Post hoc testing (a deviation from the original protocol) revealed no relationship between the presence of injury on ultrasound assessment on an unaffected leg and any other variable ($p > 0.27$).

TPD thresholds were no different between legs for Patients ($Z = -1.30$, $p = 0.19$), nor between Patients and Controls ($U = 89$, $p = 0.70$). Patients were less accurate at localising tactile stimuli on their affected leg than on their unaffected leg ($Z = -2.98$, $p = 0.004$), and the matched affected leg of Controls ($U = 42.5$, $p = 0.009$) (Fig. 1A).

Patients were slower for left/right judgements when lower limb images corresponded to their affected leg than when they corresponded to their unaffected leg ($t(13) = 4.41$, $p = 0.001$), or the matched affected leg of Controls ($t(26) = 2.88$, $p = 0.008$) (Fig. 1B). There were no within group ($t(13) = 0.00$, $p = 1.00$) or between group ($t(26)$ 0.03, $p = 0.97$) differences for left/right judgements of upper limb images.

For the auditory stimulus detection task, Controls scored 100% accuracy on both legs. Patients scored 100% on the unaffected leg and 86.4 ± 17.6% on the affected leg. While visual inspection of the data indicates a difference between the affected and unaffected legs in Patients (Fig. 2), the 100% scores and lack of variance on the Patient's unaffected leg and on both legs of Controls rendered Wilcoxon signed rank test unsuitable.

4. Discussion

Leg-specific tactile, proprioceptive, and spatial processing deficits exist in these elite athletes with persistent posterior thigh pain. These deficits likely reflect a disruption in cortical representations. That they exist in these athletes despite rehabilitation and normal tissue healing time, raises the possibility (but does not

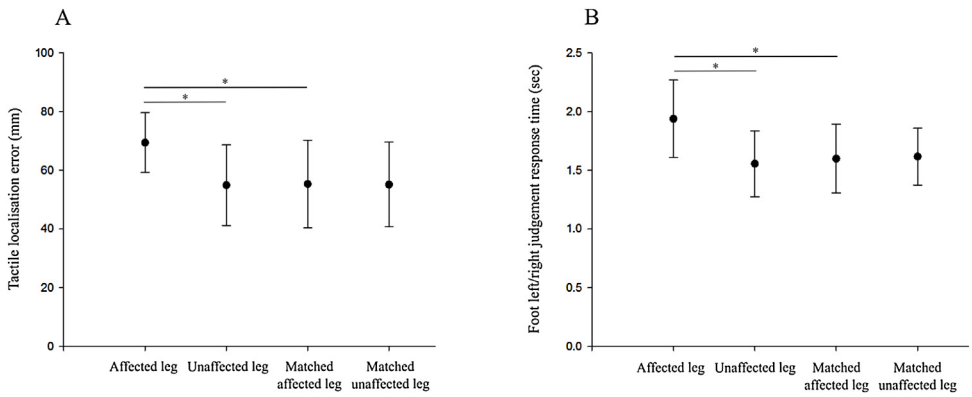

Fig. 1. Mean and standard deviation for tactile localisation error (A) and response times for the left/right judgement of foot images for Patients (affected and unaffected sides) and Controls (matched affected and matched unaffected sides). $^{*}p < 0.05$.

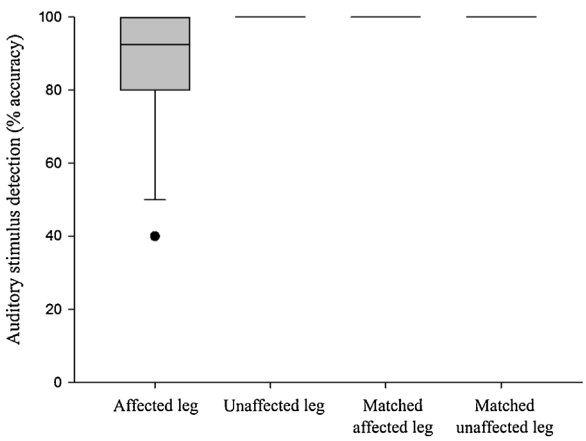

Fig. 2. Median and interquartile ranges for the auditory detection task for Patients (affected and unaffected sides) and Controls (matched affected and matched unaffected sides). Note the 100% accuracy of detecting the auditory cue when delivered to the unaffected leg of Patients, and on either leg of Controls.

prove) that this disruption in cortical representations may contribute to the persistence of posterior thigh pain.

To our knowledge, this is the first report of atopognosia (poor tactile localisation) in an elite athletic population with persistent musculoskeletal pain, but there are several accounts from non-athlete populations, including persistent LBP,[47] phantom limb pain (PLP),[26] and CRPS.[29] However, TPD thresholds were not different between Patients and Controls, suggesting that impaired tactile function is specific to localisation rather than discrimination of stimuli. Evidence suggests that mislocalisation of tactile stimuli may be related to the reorganisation of representations within the primary somatosensory cortex (S1),[29] while discrimination of tactile stimuli is related to the receptive field size of the representation.[20] TPD is actually more dependent on intensity cues than spatial cues,[11] so atopognosia without TPD increase may reflect reorganisation but not altered receptive field size, within S1.

Patients were slower than Controls at the left/right lower limb judgement task when the image coincided with their affected side. Again, similar side-specific deficits have been observed in other persistent pain conditions including CRPS, PLP, and carpal tunnel syndrome.[5] Left/right judgements require two steps: first, an initial left/right selection is made, and second, the cortical proprioceptive map is accessed and the body is mentally rotated into a position that matches the one depicted in the image to confirm or refute the initial selection. Longer response times in the left/right judgement task are thought to reflect a bias towards the unaffected leg when accessing the cortical proprioceptive map, generating a refuted decision during the mental rotation, thereby causing a delay in the final response as the mental rotation process is repeated for the other side (see[5] for more discussion). Importantly, such deficits have been suggested to contribute to the persistence of pain in some conditions.[31] However, whether the observed deficits in left/right judgement performance underlie the persistence of posterior thigh pain cannot be determined from this study and requires further investigation.

Patients were less accurate than controls at detecting an auditory cue delivered near their affected leg. While previous studies have assessed auditory stimulus detection in neurological populations (e.g., visuospatial neglect patients[43]), except a facial pain case study,[32] it appears unexplored in musculoskeletal pain. That Patients were less likely to detect an auditory cue delivered near their affected leg suggests a general decrease in weighting (or influence) of cues arising from the space surrounding the affected leg. This finding seems consistent with the bias in left/right judgement performance observed here, and with the spatially-defined deficits observed in CRPS (e.g., patients with CRPS have been shown to prioritise tactile information from the space where the unaffected limb resides over that of where the affected limb resides, and have demonstrated motor deficits that are biased towards the unaffected side of space[37,44]). This side-to-side imbalance in sensory/motor processing in CRPS has been termed 'somatospatial neglect'.[44] The data presented here suggests that similar somatospatial neglect may exist in elite athletes with persistent posterior thigh pain. Understanding the relevance of such spatial deficits for the recovery, and risk of re-injury, following an acute hamstring strain is warranted.

The current results are of potential importance because they raise the possibility that rehabilitation after hamstring strains might warrant training of the cortical representations of the painful body part. This approach has been used in other persistent pain conditions and shown promising results, with evidence most established for sensory discrimination training[38] and graded motor imagery (GMI).[4] There is good evidence that GMI and its components activate the sensorimotor cortex[14] and reduce pain in CRPS and PLP,[35] with early evidence of its effectiveness in persistent LBP.[12] Although these clinical data provide support for the integration of cortical representation theory and training in sports rehabilitation, little data exists of its effectiveness within a sporting context and this warrants further investigation.

One incidental finding is worthy of note. Of 26 unaffected hamstrings assessed on ultrasound by a blinded assessor, eight were diagnosed with a Grade 1 injury ($n = 4$ uninjured side of patients, $n = 4$ asymptomatic controls). There was no difference between

these eight individuals and the remainder of the group on any variables assessed on the unaffected hamstring. Given that ultrasound has poor specificity in determining Grade 1 hamstring tears,[25] it is possible that these Grade 1 tears were false positives; that is, they provided an ultrasound diagnosis of a Grade 1 hamstring tear without any tissue damage present. If these Grade 1 tears were not false positives but indeed were true minor injuries that were never symptomatic, it would add to existing literature that differentiates pain and tissue damage; that is, that tissue damage is neither necessary nor sufficient for an experience of pain.[39] There is evidence from several other populations of positive investigations without symptoms, for example, in asymptomatic individuals with evidence of abnormal MRI scans of the lumbar spine[2] and knee.[19]

This study has limitations. Left/right judgements, tactile acuity tasks, and auditory stimulus detections tasks are clearly not direct assessments of cortical representations; each depends on the integrity of processes at several levels of the neuraxis. However, there is a solid basis for interpreting disrupted performance on these tasks (presuming testing is carried out in a manner that controls for peripheral disruptions, as was performed here) as reflecting disrupted cortical processes.[33] Assessments that might corroborate that interpretation include neurophysiological or hemodynamic brain imaging techniques. Without them, our confidence in attributing disrupted performance to disrupted cortical representations should be appropriately tempered. A small convenience sample reflects our intent to investigate these issues in very elite performers, but it means that the findings may not generalise to the wider athletic population. Given the cross-sectional nature of the study, causal relationships cannot be demonstrated. Thus, it is not known whether disruptions in cortical representations are a cause or a consequence of posterior thigh pain, or if they are both influenced by an unmeasured confounder. Longitudinal studies are required to better understand the causal nature of this relationship. This study has strengths too. A credible control group was used in the present study (i.e., age, gender, performance level and BMI matched), which mitigated the potential confounding effects of those factors. A prospective analysis plan was lodged, and analysis was blinded (although data collection was almost complete well before the field endorsed practices such as lodging full protocols before commencement[27] – we acknowledge that not doing that here limits transparency). Lastly, this study introduced a novel assessment of spatial processing of auditory stimuli and included reliability testing. Future work should explore this assessment in other persistent pain conditions and determine how deficits in spatial processing relate to clinical outcomes.

5. Conclusions

Leg-specific tactile, proprioceptive, and spatial deficits exist in athletes with persistent posterior thigh pain, and such deficits suggest disruptions in various cortical representations. That these deficits exist despite intensive rehabilitation and normal tissue healing time, suggests that they may be related to the persistence of pain. Further work is required to better understand how these deficits impact recovery following hamstring injury, and risk of re-injury.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.jsams.2020.07.003.

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